

IT/CS 311 Final Laboratory Activity

Docker Containerization

Due: 29 November 2025

Objectives. The team must be able to:

- Utilize docker containerization (multi-container) for web application deployment.
- Develop/enhance group dynamics.
- Enhance presentation skills.

Requirements:

Each team will have to deploy the Node.js and PHP web application projects (web app class requirement) in a Docker container. For this activity, you will have to create the files

Dockerfile.nodejs and **Dockerfile.apache** which will reference an image of Node and Apache respectively. These files should contain the commands to customize the images, e.g., copy the necessary files (ex. package*.js) and install the required dependencies of the Node.js and PHP projects.

Once the dockerfiles have been created, create the **compose.yaml** file that will specify the three services: **Dockerfile.nodejs**, **Dockerfile.apache**, and the **MySQL** database. Ensure that the project folders (ex. source code files in Node.js) are properly mapped to the appropriate folder/directory of the container to be created.

If the web app projects in Node.js/PHP are not yet complete, you may opt to submit your application with minimal features. Just ensure that your source codes (Node.js and PHP) will have **at least read access to the database** that is found in another container.

Deliverables:

1. Peer Evaluation - Use the template provided during prelim and midterm.

Naming convention: **ClassCode-Team#-PeerEval.pdf**

where **#** is the team's assigned number

Example: **9300-Team1-PeerEval.pdf**

2. Video recording

This video shall cover a discussion for both the **docker** and **compose.yaml** files as well as a demonstration of how to execute the docker project. Note further that the language to be used in the presentation should be purely English. Maximum duration for the video: **10 minutes**.

Naming convention: **ClassCode-Team#-Video.mp4**

where **#** is the team's assigned number

Example: **9300-Team1-Video.mp4**

3. Docker project

Archived form of the entire docker project (folders/directories) containing the required files (source code file/s, docker file/s, yaml, etc.).

Naming convention of the docker project folder: **ClassCode-Team#-Project**

where # is the team's assigned number Example: **9300-Team1-Project.zip**

Rating Scale:

5 - Excellent 4 - Very Good 3 - Adequate 2 - Needs Improvement 1 - Very Poor

Criteria:

Group score will be rated as follows:

- 50%: Completeness of the demonstrated tasks (3 services: 2 web app services, database service)
- 40%: Accuracy of the files (docker and yaml) generated to perform the tasks
- 10%: Compliance to the prescribed language/file naming conventions

Individual scores will be based on the group score adjusted using the peer evaluation.